Introduction

This project investigates patterns in student submissions using statistical analysis and machine learning techniques. Two hypotheses were examined. The first hypothesis explores whether the number of submission attempts is related to the likelihood of success in exercises, as analyzing retry behavior can help identify students who exhibit exploratory, trial-and-error, or repetitive submission behaviors. The second hypothesis examines whether the timing of submissions affects success rates. Since exam submissions occur under controlled conditions, the analysis primarily focuses on laboratory submissions to better capture natural student working behavior.

For Hypothesis 1:

- H0: There is no relationship between the number of submission attempts and all-test-case success.
- H1: There is a significant relationship between the number of submission attempts and all-test-case success.

For Hypothesis 2:

- H0: The all-test-case pass rate during peak hours is equal to the pass rate outside peak hours.
- H1: The all-test-case pass rate during peak hours is significantly different from the pass rate outside peak hours.

Data Collection and Preparation

The dataset used in this study consists of submissions collected from ITCS113 and ITDS120 laboratory and examination activities in JSONL format. A custom extraction function was developed to transform the raw files into a structured tabular dataset. Since one submission may contain several test cases, each test case was initially recorded as a separate row.

Several preprocessing procedures were performed on the dataset:

- Unreliable records such as SYSTEM_ERROR were removed
- Test result labels were standardized into consistent categories
- Timestamps were converted into datetime format
- Test-case-level data was aggregated into submission-level data using message_id

The final cleaned dataset contained:

- submission information
- activity type
- exercise identifiers
- timestamps
- test case outcomes
- retry behavior

Methodology

Hypothesis 1

To analyze retry behavior, the dataset was transformed from test-case level into submission-level data. Each submission was grouped using message_id. The number of test cases was calculated as num_cases, while a submission was considered successful only when all test cases passed successfully. Submissions were sorted according to user, exercise, and submission time. Attempt numbers were then generated sequentially for each student-exercise pair using cumulative counting. The success rate for every attempt number was calculated by taking the mean value of the boolean is_success variable. Since attempt numbers contained several extreme outliers, the IQR method was applied to determine an upper bound for analysis and visualization.

Normality was evaluated using the Shapiro-Wilk test. Because the data was not normally distributed, the Mann-Whitney U test was selected to compare attempt distributions between successful and unsuccessful

submissions. The statistical test produced a p-value smaller than 0.05, resulting in the rejection of the null hypothesis. This result indicates that a statistically significant relationship exists between retry behavior and submission success.

Findings and Insights

The results showed that success rates generally decreased as the number of attempts increased. Students who completed exercises with fewer attempts tended to achieve higher success rates, whereas students who required repeated retries were less likely to succeed. These findings suggest that repeated attempts may indicate difficulties in understanding problems or debugging code effectively. At extremely high attempt numbers, success rates became unstable because of the small sample size. Therefore, the interpretation mainly focused on early and medium-range attempts.

Hypothesis 2

The second hypothesis investigated whether submission timing influences success rates. Only laboratory submissions were included in the analysis because examination submissions occur during controlled schedules. Submission hours were extracted from creation_time, and the dataset was divided into the following groups:

- Peak hours: 06:00–11:59
- Non-peak hours: all remaining hours

A violin plot was used to compare the distribution of submission hours between successful and unsuccessful submissions. The Shapiro-Wilk test showed that the pass rates were not normally distributed. Levene's test indicated that the variance between groups was similar. Therefore, the Mann-Whitney U test was applied to compare pass-rate distributions between peak and non-peak hours. The statistical test produced a p-value below 0.05, leading to the rejection of the null hypothesis.

Findings and Insights

Although submission activity changed throughout different times of the day, the overall success distributions between peak and non-peak hours remained relatively similar. The findings suggest that submission timing has only a limited effect on success rates when compared with stronger behavioral factors such as retry behavior. Students were able to achieve successful submissions across different time periods, indicating that the time of day alone may not strongly influence performance.

Machine Learning

To further extend Hypothesis 1, a machine learning model was developed to predict retry behavior based on submission characteristics. The dataset was first aggregated into submission-level records using message_id. Submission success rate was calculated as the number of successful test cases divided by the total number of test cases within a thread id. Attempt numbers were generated by sorting submissions according to student, exercise, and submission time, followed by cumulative counting within each student-exercise group. Extreme retry behavior was removed using the previously calculated IQR upper bound. Additional temporal features, including submission hour and day of week, were extracted from creation_time.

The attempt number was transformed into a binary retry category using a threshold of 10 attempts:

- 1 = within 10 attempts
- 0 = above 10 attempts

The activity_type feature was numerically encoded as follows:

- exam = 1
- lab = 0

The final feature set included:

- activity type

- success rate
- submission hour
- number of test cases
- day of week

A Logistic Regression model with polynomial feature expansion and balanced class weighting was trained using an 80:20 train-test split. Five-fold cross-validation was applied to evaluate model stability, while class_weight='balanced' was used to address the imbalanced dataset.

The model achieved the following results:

- Accuracy ≈ 0.64
- Precision ≈ 0.69
- Recall ≈ 0.67
- F1-score ≈ 0.68

The training and validation MSE values were relatively close, indicating moderate generalization performance with slight overfitting. The findings suggest that submission behavior features contain meaningful information related to retry behavior, although the prediction performance remains moderate. These results indicate that learning analytics may help identify students who are more likely to require repeated submission attempts.

Practical Implication

With further improvement in model performance, instructors may be able to predict the likely number of student attempts will be above or below 10 attempts based on features such as success rate, activity type, submission time, number of test cases, and day of week. This information may help teachers determine suitable attempt limits and improve student learning outcomes.

Conclusion

This study investigated behavioral patterns in student submissions using statistical analysis and machine learning techniques. The first hypothesis demonstrated a significant relationship between retry behavior and submission success. Students who required repeated attempts generally achieved lower success rates, suggesting that repeated retries may indicate greater learning difficulty.

The second hypothesis examined whether submission timing affects success rates. Although submission behavior varied across different hours, the findings suggest that timing alone has only a limited influence on success.

The machine learning analysis further demonstrated that submission behavior features can moderately predict retry patterns in exercises.

Overall, the findings indicate that behavioral indicators such as retry behavior and submission patterns can provide meaningful insights into student learning behavior. These insights may help instructors identify students earlier and improve educational support strategies.

Appendix

```
df_submission = (
    df.groupby("message_id")
    .agg({
        "creation_time": "first",
        "thread_id": "first",
        "user_id": "first",
        "exercise_id": "first",
        "activity_type": "first",
        "test_result": lambda x: (x == "SUCCESS").all(),  # all test cases passed
        "case_no": "count"  # number of test cases
    })
    .rename(columns={
        "test_result": "is_success",
        "case_no": "num_cases"
    })
    .reset_index()
)
```

We grouped the df_submission for hypothesis1with message_id. In the case, we utilized “first” because rows of same exercise id are always same. Test_result will produce Success when all submission passed.

```
#create attempt number
df_submission = df_submission.sort_values(
    ['user_id', 'exercise_id', 'creation_time']
)
df_submission["attempt_no"] = (
    df_submission.groupby(["user_id", "exercise_id"])
    .cumcount() + 1
)
```

We sorted df_submission by user_id, exercise_id and creation_time. After that we add attempt frequency column into this table.

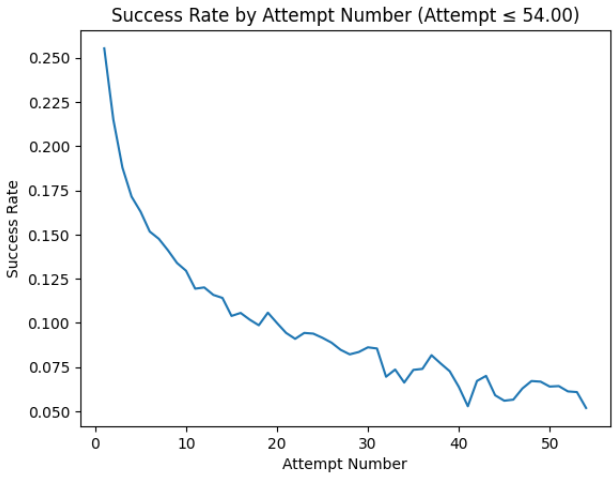
```
df_submission.groupby("attempt_no")["is_success"].mean()
```

The mean represents the proportion of successful submissions at each attempt number.

```
upper_bound = Q3 + 2 * IQR


print("Lower bound:", lower_bound)
```

Using IQR multiplier with 2.



This visualization shows that higher the attempt, lower the success.

```
from scipy.stats import shapiro
data = df_plot['attempt_no']
stat, p = shapiro(data)
print('stat=%.3f, p=%.3f' % (stat, p))
if p > 0.05:
print('Probably normally distributed' )
else:
print('Probably not normally distributed' )
```

```
stat=0.841, p=0.000
Probably not normally distributed
```

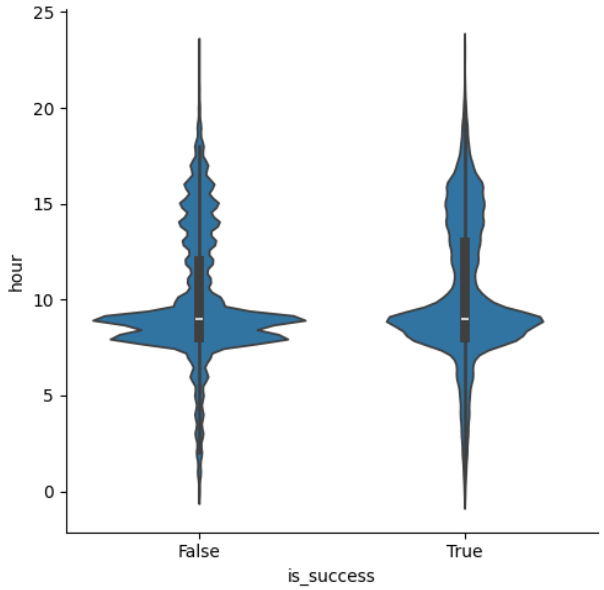
By using shapiro, deciding normal or non-normal distribution.

```
from scipy.stats import mannwhitneyu
```

Mann-Whitney U test is because of non-normal distribution.

```
data1 = df_plot[df_plot["is_success"] ==
True]["attempt_no"]
data2 = df_plot[df_plot["is_success"] ==
False]["attempt_no"]
```

For testing hypothesis1, spiriting two group data1(success) and data2(unsuccess) by attempt number.



This image shows nearly same distribution between records that passed all test and records that are not.

```
df_group = (
    df_lab.groupby("hour")
    .agg({
        "is_success": ["mean", "count",
"sum"]
    }).rename(columns={
        "mean": "pass_rate",
```

```
        "count": "total_count",
        "sum": "all passes"
    })
    .reset_index()
)
```

Creation required columns for Hypothesis2 grouping with hour by applying “mean”, “count” and “sum” to is_success.

```
def time_bin(h):
    if ((h>=6) & (h < 12)) :
```

Defining peak hour (6am to 12 am) and out of that range is not peak hour.

```
data1 = df_group[df_group['time_of_day'] ==
'peak hours']['pass_rate']
data2 = df_group[df_group['time_of_day'] ==
'Not peak hours']['pass_rate']
```

For testing hypothesis2, spiriting two group data1(peak hours) and data2(Not peak hours) by pass rate.

```
"test_result": lambda x: (x ==
"SUCCESS").sum() / len(x)
```

Calculation of success rate for each submission.

```
df_ml["attempt_no"] = (
  df_ml.groupby(["user_id", "exercise_id"])
  .cumcount() + 1
)
```

The attempt number was generated using cumulative counting.

```
df_ml = df_ml[df_ml["attempt_no"] <=
upper_bound]
```

To avoid extreme, retry behavior we use the previous calculated upper bound.

```
df_ml = df_ml[['creation_time','activity_type',
'attempt_no','num_cases','success_rate']]
```

For feature selection, we only select relevant columns for machine learning

```
df_ml["hour"] = df_ml["creation_time"].dt.hour
df_ml["dow"] =
df_ml["creation_time"].dt.dayofweek
```

For time-based feature extraction, we extract additional temporal features from creation_time.

```
def att_rg(h):
    if h <= 10:
        return 1
    else:
        return 0

df_ml["attempt_no"] =
df_ml["attempt_no"].apply(att_rg)
```

After that we transformed attempt number into binary retry category

```
pipe = make_pipeline(
    StandardScaler(),     # Mean-var scaler
    #RobustScaler(),     # IQR Scaler
    PolynomialFeatures(degree=2),
    #LinearRegression(),   # No
regularization
    LogisticRegression(max_iter=100000,
class_weight='balanced')
```

We created a machine learning pipeline with StandardScaler, PolynomialFeatures(degree=2) and LogisticRegression.